MAT2040

Tutorial 7

CUHK(SZ)

June 26, 2024

Question 1

Suppose $A = [a_1, a_2, a_3, a_4, a_5, a_6, a_7]$, and U is the reduced row echelon form of A, where

$$U = \begin{bmatrix} 1 & 0 & 2 & 0 & 1 & 0 & 3 \\ 0 & 1 & -1 & 0 & 2 & 0 & 1 \\ 0 & 0 & 0 & 1 & 3 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 \end{bmatrix}$$

Find a basis for the null space, column space and row space of A, respectively.

Question 1 - Solution

Solving the equation $A\mathbf{x} = \mathbf{0}$ gives:

$$\begin{cases} x_1 = -2x_3 - x_5 - 3x_7 \\ x_2 = x_3 - 2x_5 - x_7 \\ x_3 = x_3 \text{ (free variable)} \\ x_4 = -3x_5 - 2x_7 \\ x_5 = x_5 \text{ (free variable)} \\ x_6 = x_7 \\ x_7 = x_7 \text{ (free variable)} \end{cases}$$

Therefore a basis of null space for A can be:

$$\left\{ \begin{pmatrix} -2 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ -2 \\ 0 \\ -3 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -3 \\ -1 \\ 0 \\ -2 \\ 0 \\ 1 \\ 1 \end{pmatrix} \right\}$$

Question 1 - Solution

A basis of column space can be formed by columns of A that corresponding to the pivot columns of U. The pivot columns of U are 1st, 2nd, 4th, and 6th columns, therefore a basis of column space of A can be:

$$\{a_1, a_2, a_4, a_6\}$$

Because row operation won't change the row space, the row space of A should be equal to the row space of U. Note that the four row vectors in U are linearly independent, so they form a basis of row space of A :

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ 2 \\ 0 \\ 1 \\ 0 \\ 3 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \\ 2 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 3 \\ 0 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ -1 \end{pmatrix} \right\}$$

Question 2

Let

$$A = \begin{bmatrix} 1 & -3 & 2 & 0 \\ 2 & 1 & 0 & -1 \\ 4 & 8 & -4 & -3 \\ 1 & 10 & -6 & -2 \end{bmatrix}$$

(1) Transform A to RREF, then deduce a basis of the row space of A.

(2) Calculate the dimension of the column space of A, and the nullity of A.

(3) Find a basis of the row space of A consisting entirely of row vectors of A.

Question 2 - Solution

1. To obtain the reduced row echelon form (RREF) of the matrix A,

$$\begin{bmatrix} 1 & -3 & 2 & 0 \\ 2 & 1 & 0 & -1 \\ 4 & 8 & -4 & -3 \\ 1 & 10 & -6 & -2 \end{bmatrix} \xrightarrow[R_4 \to R_4 - R_1]{\substack{R_2 \to R_2 - 2R_1 \\ R_3 \to R_3 - 4R_1}} \begin{bmatrix} 1 & -3 & 2 & 0 \\ 0 & 7 & -4 & -1 \\ 0 & 20 & -12 & -3 \\ 0 & 13 & -8 & -2 \end{bmatrix} \xrightarrow[R_4 \to R_4 - \frac{13}{7}R_2]{R_3 \to R_3 - \frac{20}{17}R_2}$$

$$\begin{bmatrix} 1 & -3 & 2 & 0 \\ 0 & 7 & -4 & -1 \\ 0 & 0 & -\frac{4}{7} & -\frac{1}{7} \\ 0 & 0 & -\frac{4}{7} & -\frac{1}{7} \end{bmatrix} \xrightarrow[R_2 \to \frac{1}{7}R_2]{\substack{R_4 \to R_4 - R_3 \\ R_3 \to -\frac{1}{4}R_3}} \begin{bmatrix} 1 & -3 & 2 & 0 \\ 0 & 1 & -\frac{4}{7} & -\frac{1}{7} \\ 0 & 0 & 1 & \frac{1}{4} \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[R_1 \to R_1 - 2R_3]{R_2 \to R_2 + \frac{4}{7}R_3}$$

$$\begin{bmatrix} 1 & -3 & 0 & -\frac{1}{2} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{4} \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \to R_+ 3R_2} \begin{bmatrix} 1 & 0 & 0 & -\frac{1}{2} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{4} \\ 0 & 0 & 0 & 0 \end{bmatrix} = \mathrm{RREF}(A)$$

The non-zero rows of $\mathrm{RREF}(A)$ constitute a basis of the row space of A. Thus, $\left\{ \left(1, 0, 0, -\frac{1}{2}\right)^{\mathrm{T}}, (0, 1, 0, 0)^{\mathrm{T}}, \left(0, 0, 1, \frac{1}{4}\right)^{\mathrm{T}} \right\}$ is a basis of $\mathrm{Row}(A)$.

Question 2 - Solution

2.
We know that $\dim \mathrm{Col}(A) = \dim \mathrm{Row}(A) = \mathrm{rank}(A)$.
From above, $\dim \mathrm{Row}(A) = 3$; thus $\mathrm{rank}(A) = 3$.
A is a 4×4 matrix; by Rank-Nullity Theorem, $\mathrm{rank}(A) + \mathrm{nullity}(A) = 4$; thus, $\mathrm{nullity}(A) = 4 - 3 = 1$.

3.
Let $B = A^T = \begin{bmatrix} 1 & 2 & 4 & 1 \\ -3 & 1 & 8 & 10 \\ 2 & 0 & -4 & -6 \\ 0 & -1 & -3 & -2 \end{bmatrix}$. Then, $\mathrm{Row}(A) = \mathrm{Col}(B)$.
Transforming B to RREF, we obtain $C = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.
The columns of B corresponding to the pivot columns of C constitute a basis of $\mathrm{Col}(B)$.
Thus, $\left\{\mathbf{r}_1 := (1, -3, 2, 0)^{\mathrm{T}}, \mathbf{r}_2 = (2, 1, 0, -1)^{\mathrm{T}}, \mathbf{r}_3 := (4, 8, -4, -3)^{\mathrm{T}}\right\}$ is a basis of $\mathrm{Row}(A)$.

Question 3

Let $A, B \in \mathbb{R}^{n\times n}$. Show that $\text{rank}(A+B) \leq \text{rank}(A) + \text{rank}(B)$.

Solution

Suppose that $A = [a_1, \ldots, a_n]$ and $B = [b_1, \ldots, b_n]$.
$A + B = [a_1 + b_1, \ldots, a_n + b_n]$.

Suppose the pivot columns of A are $a_{i_1}, \ldots, a_{i_k}$ and the pivot columns of B are $b_{j_1}, \ldots, b_{j_s}$, then $a_{i_1}, \ldots, a_{i_k}$ is a basis for $\mathrm{Col}(A)$ and $b_{j_1}, \ldots, b_{j_s}$ is a basis for $\mathrm{Col}(B)$.

Moreover, it is easy to verify that
$\mathrm{Span}\,(a_{i_1}, \ldots, a_{i_k}, b_{j_1}, \ldots, b_{j_s}) = \mathrm{Span}\,(a_1, \ldots, a_n, b_1, \ldots, b_n)$ and
$\mathrm{Col}(A + B) = \mathrm{Span}\,(a_1 + b_1, \ldots, a_n + b_n) \subseteq \mathrm{Span}\,(a_1, \ldots, a_n, b_1, \ldots, b_n)$.
Thus, $\dim(\mathrm{Col}(A + B)) \leq \dim\,(\mathrm{Span}\,(a_{i_1}, \ldots, a_{i_k}, b_{j_1}, \ldots, b_{j_s})) \leq k + s$.

Since $\mathrm{rank}(A) = k$, $\mathrm{rank}(B) = s$, one has $\mathrm{rank}(A + B) \leq$
$\mathrm{rank}(A) + \mathrm{rank}(B)$, where $\mathrm{rank}(A + B) = \dim(\mathrm{Col}(A + B))$ is used.

Question 4

Given a matrix $A \in \mathbb{R}^{m \times n}$ with $\text{rank}(A) = r$. Suppose there exist $\boldsymbol{b} \in \mathbb{R}^m$ for which $A\boldsymbol{x} = \boldsymbol{b}$ has no solution.

(a) What are the inequalities ($<$ or $\leq$) that must be satisfied by m, n and r ?

(b) Does $A^T \boldsymbol{y} = \mathbf{0}$ have any other solutions besides $\boldsymbol{y} = \mathbf{0}$?

Solution

(a) The rank of a matrix is always less than or equal to the number of rows and columns, so $r \leq m$ and $r \leq n$.
Because there exist $\boldsymbol{b}$ such that $A\boldsymbol{x} = \boldsymbol{b}$ has no solution, $\boldsymbol{b}$ is not in the column space of A. we can infer that $\text{Col}(A) \subsetneq \mathbb{R}^m$, so $r < m$.

(b) The solution of $A^T\boldsymbol{y} = \boldsymbol{0}$ makes up Null $\left(A^T\right)$, $A^T \in \mathbb{R}^{n\times m}$.
Nullity $\left(A^T\right) = m - r$, and $m - r > 0$.
Therefore, there exists non-zero $\boldsymbol{y} \in \mathbb{R}^m$ satisfying $A^T\boldsymbol{y} = \boldsymbol{0}$.